WORKED SOLUTION

STEP II 1995, Question 4

Pure • Wallis Reduction Formula

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 4.4, *Wallis, Beta and the Factorial of a Half*: https://beyondstep.keithmcnulty.org/ch04/wallis-beta.html

The key idea

Integrating by parts turns $\cos^2 t$ into $1 - \sin^2 t$, which gives a reduction formula linking u_n to u_{n-2}. Multiplying that formula by u_{n-1} shows that nu_nu_{n-1} never changes as n increases, so it equals its value at $n = 1$. After that the question is a squeeze: since the u_n decrease, the product u_nu_{n-1} lies between u_n^2 and u_{n-1}^2.

Solution

The reduction formula

Let $n \geq 2$. We write $u_n = \int_0^{\pi/2} \sin t \, \sin^{n-1} t \, \mathrm{d}t$ and integrate by parts, differentiating $\sin^{n-1} t$ and integrating $\sin t$ to $-\cos t$:

$$\begin{aligned} u_n &= \Big[-\cos t \, \sin^{n-1} t\Big]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \cos^2 t \, \sin^{n-2} t \, \mathrm{d}t \\ &= 0 + (n-1)\int_0^{\pi/2} (1 - \sin^2 t)\sin^{n-2} t \, \mathrm{d}t \\ &= (n-1)(u_{n-2} - u_n). \end{aligned}$$

The boundary term vanishes because $\cos\frac{\pi}{2} = 0$ at the top and $\sin 0 = 0$ at the bottom (here we use $n - 1 \geq 1$). Collecting the u_n terms gives

$$nu_n = (n-1)u_{n-2} \qquad (n \geq 2).$$

Multiplying both sides by u_{n-1} gives

$$nu_nu_{n-1} = (n-1)u_{n-1}u_{n-2} \qquad (n \geq 2),$$

as required.

The constant product

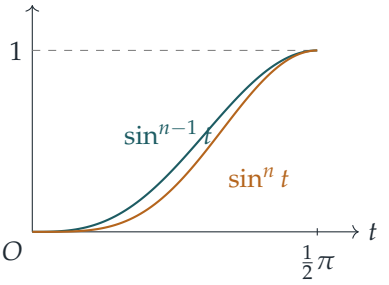
Write $p_n = nu_nu_{n-1}$ for $n \geq 1$. The last result says $p_n = p_{n-1}$ for every $n \geq 2$, so $p_n = p_1$ for every $n \geq 1$. Now

$$u_0 = \int_0^{\pi/2} 1 \, \mathrm{d}t = \tfrac{1}{2}\pi, \qquad u_1 = \int_0^{\pi/2} \sin t \, \mathrm{d}t = \Big[-\cos t\Big]_0^{\pi/2} = 1,$$

so $p_1 = 1 \cdot u_1u_0 = \frac{1}{2}\pi$. Hence

$$nu_nu_{n-1} = \tfrac{1}{2}\pi \qquad \text{for all } n \geq 1.$$

The sketch and $0 < u_n < u_{n-1}$



Both curves start at 0 when $t = 0$, increase, and reach 1 at $t = \frac{1}{2}\pi$ (the sketch shows $n = 4$; for $n = 1$ the upper curve is the line $y = 1$). For $0 < t < \frac{1}{2}\pi$ we have $0 < \sin t < 1$, so

$$0 < \sin^n t = \sin t \cdot \sin^{n-1} t < \sin^{n-1} t.$$

So the graph of $\sin^n t$ lies strictly below that of $\sin^{n-1} t$ except at the two ends. Both functions are continuous, so the areas under them satisfy

$$0 < u_n < u_{n-1} \qquad (n \geq 1).$$

Ask yourself Why are the inequalities strict, when the curves touch at the ends?

The difference $\sin^{n-1} t - \sin^n t$ is continuous, never negative, and positive at (say) $t = \frac{1}{4}\pi$. A continuous function that is positive at a point is positive on a small interval around it, so it encloses a positive area. Touching at two isolated points doesn't change that. The same argument shows $u_n > 0$.

The two-sided bound

Let $n \geq 1$. Since $u_n > 0$, multiplying $u_n < u_{n-1}$ by nu_n gives $nu_n^2 < nu_nu_{n-1}$. Since $u_{n-1} > 0$, multiplying by nu_{n-1} gives $nu_nu_{n-1} < nu_{n-1}^2$. The middle quantity is $\frac{1}{2}\pi$, so

$$nu_n^2 < \tfrac{1}{2}\pi < nu_{n-1}^2 \qquad (n \geq 1).$$

The squeeze and the limit

The right-hand inequality holds for every $n \geq 1$, so we may apply it with $n+1$ in place of n:

$$\tfrac{1}{2}\pi < (n+1)u_n^2, \qquad \text{so} \qquad nu_n^2 > \frac{n}{n+1} \cdot \tfrac{1}{2}\pi.$$

Combining this with the left-hand inequality gives

$$\left(\frac{n}{n+1}\right) \tfrac{1}{2}\pi < nu_n^2 < \tfrac{1}{2}\pi.$$

As $n \to \infty$, $\frac{n}{n+1} = 1 - \frac{1}{n+1} \to 1$, so the lower bound tends to $\frac{1}{2}\pi$. The upper bound is $\frac{1}{2}\pi$ itself. By the squeeze (sandwich) theorem,

$$nu_n^2 \to \tfrac{1}{2}\pi \quad \text{as } n \to \infty.$$

Checks

- The reduction formula gives $u_2 = \frac{1}{2}u_0 = \frac{1}{4}\pi$, which matches $\int_0^{\pi/2} \sin^2 t \, \mathrm{d}t = \frac{1}{4}\pi$ (the average of $\sin^2$ is $\frac{1}{2}$). Then $2u_2u_1 = \frac{1}{2}\pi$, as predicted.
- Numerically (Simpson's rule), $u_{10} \approx 0.38656$, which is $\frac{63}{512}\pi$. So $10u_{10}^2 \approx 1.4943$, which lies between $\frac{10}{11} \cdot \frac{1}{2}\pi \approx 1.4280$ and $\frac{1}{2}\pi \approx 1.5708$.
- Running the recursion to $n = 10^5$ gives $nu_n^2 \approx 1.570788$, already very close to $\frac{1}{2}\pi$.

Common slips

- Forgetting that the reduction formula needs $n \geq 2$, so the constant must be fixed using u_0 and u_1, not u_1 and u_2 alone.
- Claiming $u_n < u_{n-1}$ only because $\sin^n t \leq \sin^{n-1} t$. That gives $\leq$; strictness needs the inequality to be strict on an interval.
- Trying to get the lower bound from $\frac{1}{2}\pi < nu_{n-1}^2$ directly. That bounds u_{n-1}, not u_n, so shift the index first.

Beyond STEP

These are Wallis's integrals. Writing out u_{2m} and u_{2m+1} as products and using $nu_n^2 \to \frac{1}{2}\pi$ leads to Wallis's product $\frac{\pi}{2} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdots$. The same estimate fixes the constant $\sqrt{2\pi}$ in Stirling's formula $n! \sim \sqrt{2\pi n}\,(n/\mathrm{e})^n$. The squeeze argument used here is a standard tool of first-year analysis. Abbott's *Understanding Analysis* develops it carefully.

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